Group-5 · Progress Presentation

TransitBD

Mass Transport Ticketing System with Intelligent Transit Analytics

Bus · Intercity Train · Metro — one platform

Presented 3rd August 2026 • github.com/Munem3/transit-ticketing-system

1 · Problem Statement

Public transport in Bangladesh (bus, intercity train, metro) suffers from scheduling inefficiency, overcrowding, and poor seat allocation.

😟 For commuters

- Severe peak-hour congestion
- No dynamic fare transparency
- Hard to plan multi-modal transfers (metro ↔ train ↔ bus)
- No unified booking across modes

🏢 For authorities

- No prediction of peak passenger spikes
- Cannot manage congestion dynamically
- No real-time, context-aware passenger assistance
- Limited insight into demand & revenue

Our solution: a unified ticketing platform + AI-driven transit analytics.

2 · Requirement Analysis — Functional

- Secure registration & login (NextAuth.js)
- Browse routes/trips by mode with **real-time seat tracking**
- View train compartments / metro cars / bus cabins
- Transient seat holds with **countdown timers**
- Mock wallet payments — bKash / Rocket / Card
- Instant **QR-coded tickets** + receipts
- Cancellation with automatic wallet refund
- **AI** Peak Demand & Fare Predictor
- **AI** Multi-Modal Route Planner
- **AI** Customer Support Bot
- Admin: schedules, fare grid, transaction ledger

2 · Requirement Analysis — Non-Functional

🔒 Security

- bcrypt password hashing
- JWT sessions
- Server-side authorization
- Zod input validation

⚡ Performance

- Sub-second seat reads
- Indexed DB queries
- Server-rendered pages

✅ Reliability

- Transactional booking (no double-booking)
- Auto-expiry of holds

📱 Usability

- Responsive, mobile-first UI
- Clear booking flow

📈 Scalability

- Prisma ORM
- SQLite → PostgreSQL swap
- Stateless backend

🛠 Maintainability

- TypeScript end-to-end
- Modular App Router

3 · Feasibility Analysis

⚙️ Technical

- Proven stack: Next.js, Prisma, NextAuth, Gemini
- **Working prototype already built end-to-end**
- AI via Gemini API + heuristic fallback (no hard dependency)

👥 Operational

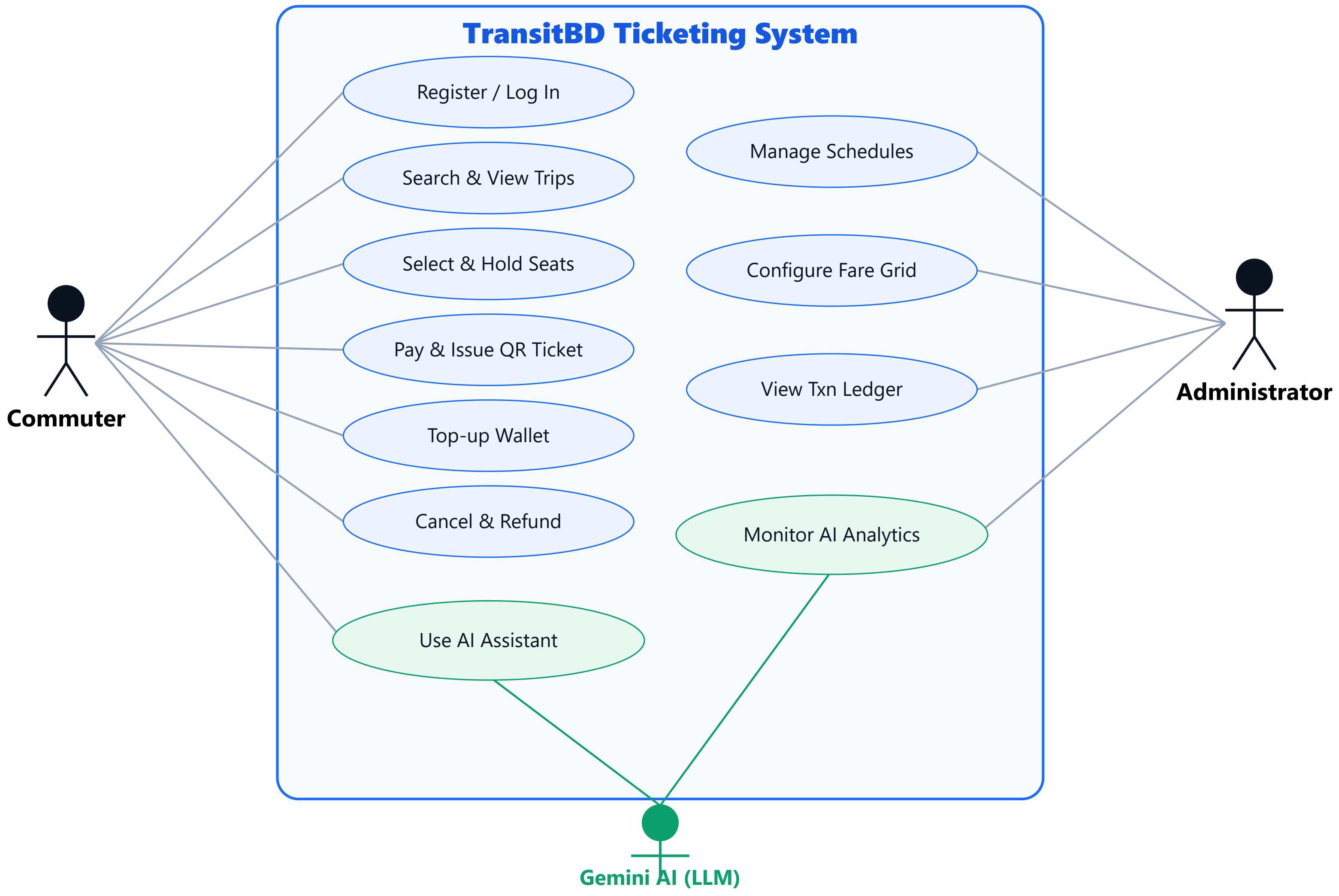
- Familiar bKash/Rocket wallet UX
- QR-at-gate already common in Dhaka Metro
- Admin panel for operators — low training need

💰 Economic

- 100% open-source stack
- Free tiers: Vercel, Gemini, Postgres
- Mock payments — no gateway fees in dev

→ Practical & achievable: the core is **already running**; remaining work is polish, live-AI tuning, and deployment.

4 · UML — Use Case Diagram



Green = AI-powered use cases (Support Bot · Route Planner · Demand & Fare Predictor)

5 · Technology Stack

Frontend

TypeScript · Next.js 14 (App Router) · React 18 · Tailwind CSS

Backend

Next.js Server Actions · API Route Handlers · Zod

Database & Auth

Prisma ORM · SQLite → PostgreSQL · NextAuth.js · bcrypt / JWT

AI & Tools

Google Gemini API · Prisma analytics · qrcode · Node.js · Git · Vercel

Analytics = Prisma aggregation over historical booking/transaction data → fed to Gemini for narrative advice.

6 · Project Timeline (3 weeks left)

Final delivery target: 24 August 2026

7 · Progress So Far

✅ Completed

- Requirement analysis & system design
- Data model — 7 entities (User, Route, Trip, Seat, Booking, Transaction…)
- Environment set up · production build passing
- Auth, route browsing, live seat maps, 5-min hold timers
- Mock wallet + transaction ledger · QR tickets · cancel/refund
- Admin panel — fare grid, ledger, trip control
- All 3 AI features (Gemini + heuristic fallback)
- Seeded data: **5 routes · 108 trips · 4,800 seats**
- Pushed to public GitHub repo

🔜 Next

- Wire live Gemini key & refine prompts
- Real-time seat refresh (polling / websockets)
- Admin analytics charts
- Automated tests & deploy to Vercel

Status: Working prototype — ahead of a typical "progress" checkpoint.

8 · Prototype Demonstration

Navigation flow:

Landing → Register / Login → Dashboard → Browse Trips → Seat Map + Hold → Pay (Wallet) → QR Ticket

Wallet / Top-up · AI Assistant (3 tabs) · Admin Panel

Built & working

- Landing, Register, Login, Dashboard
- Routes browser (mode filter, live load bars)
- Interactive seat map + hold countdown
- Wallet + ledger, Tickets + QR receipt
- AI Assistant: Support · Planner · Predictor
- Admin: stats, fare grid, ledger

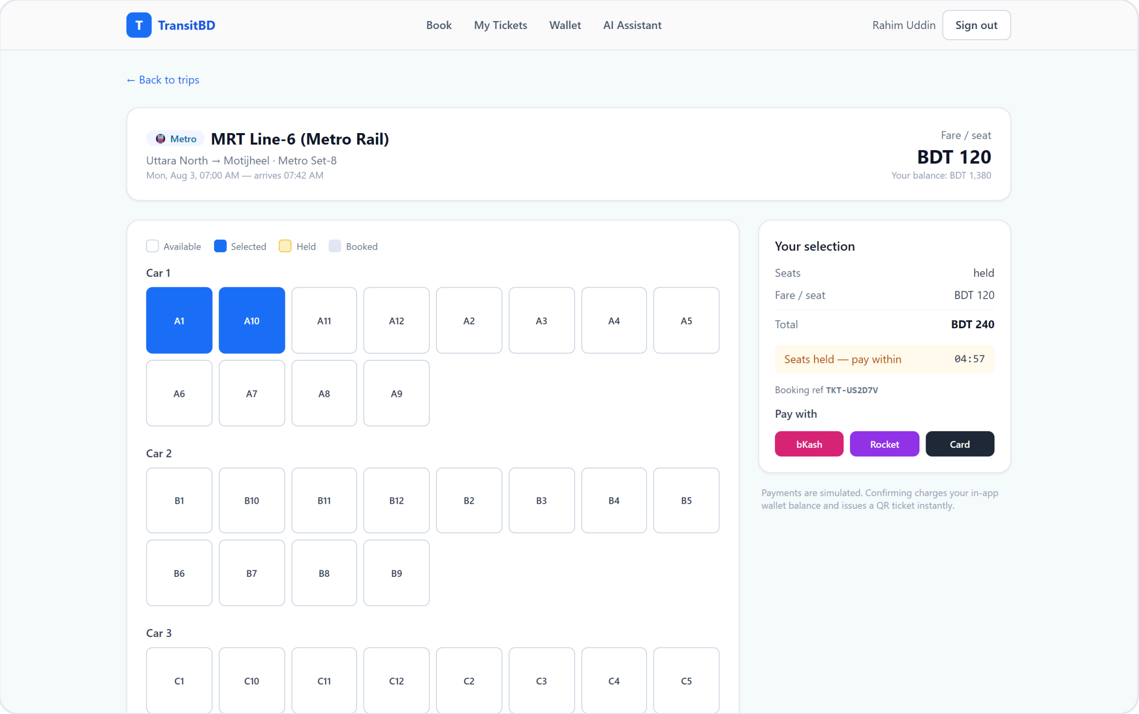
Run it yourself

- `npm run setup` → `npm run dev`
- Local: **localhost:3000**
- Admin: admin@transit.bd / admin123
- User: rahim@example.com / password123

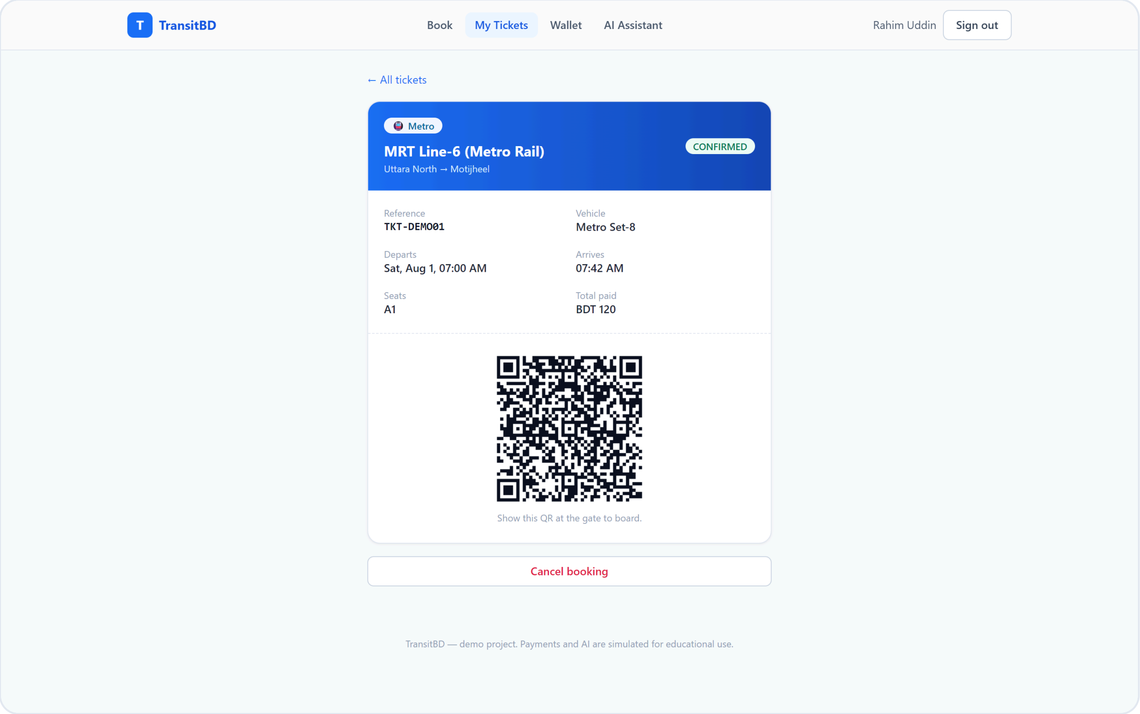
Repo: github.com/Munem3/transit-ticketing-system

Screenshots from the running prototype follow →

8 · Prototype — Booking Flow (live app)

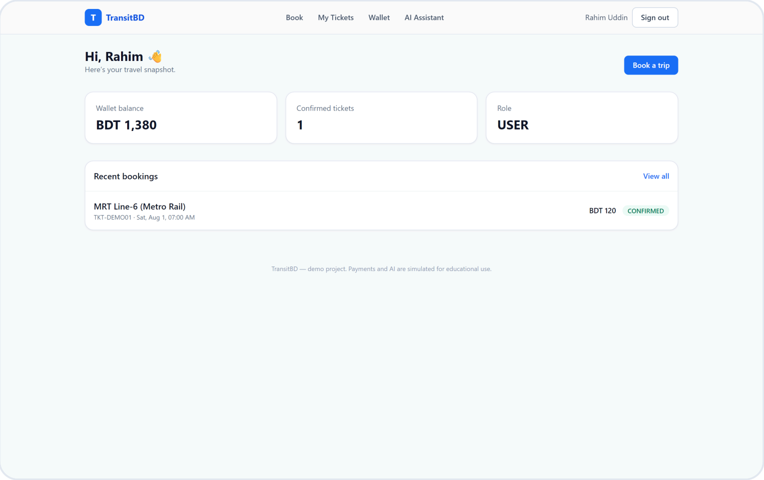


Select seats → 5-minute hold countdown → pay with bKash / Rocket / Card

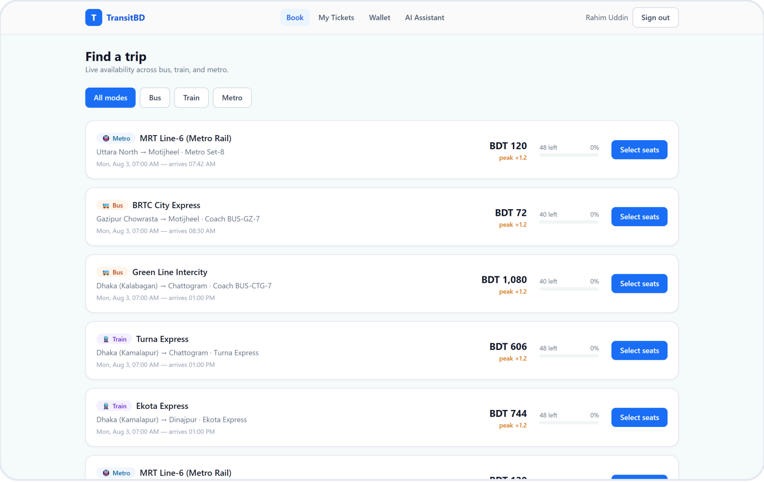


Instant QR-coded ticket issued on payment

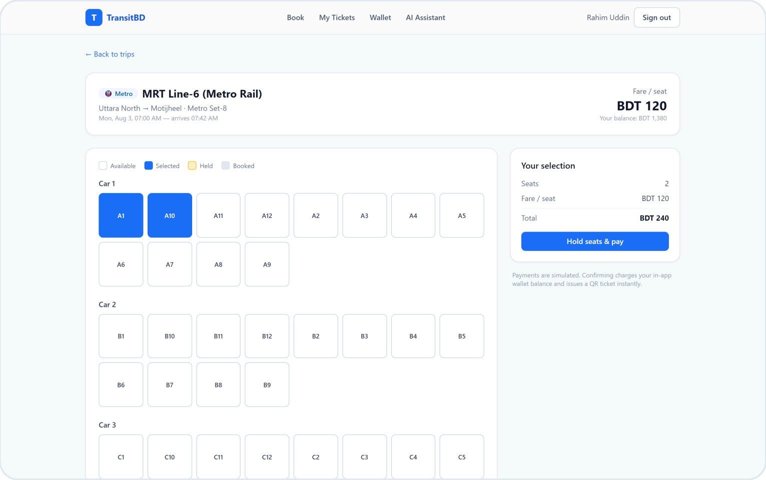
8 · Prototype — Key Screens



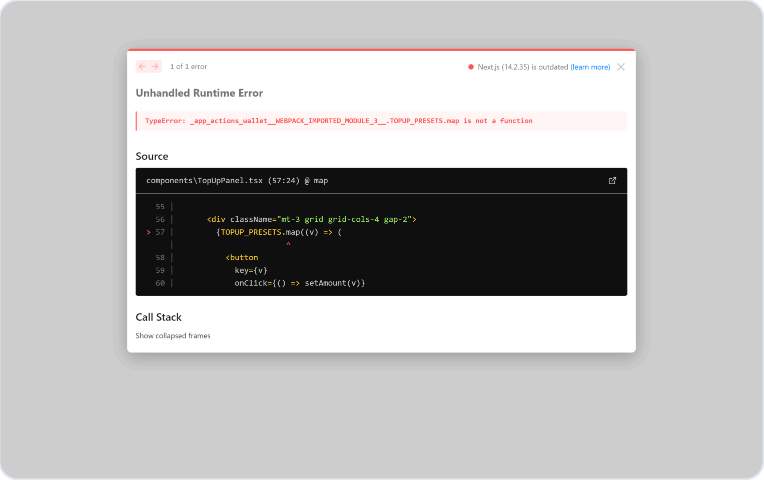
Commuter dashboard



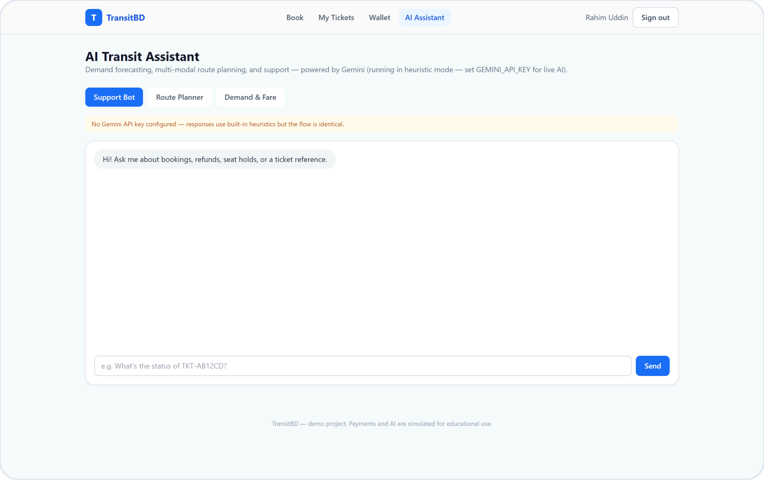
Trip browser · live seat load



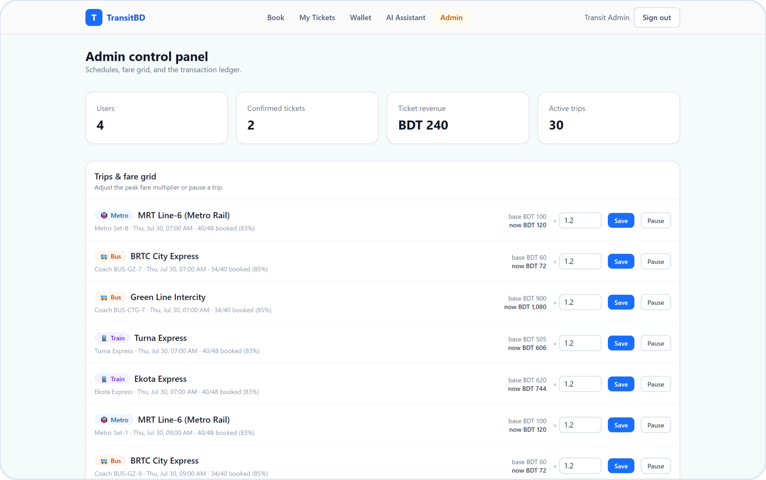
Real-time seat map



Wallet + transaction ledger



AI Assistant (3 tools)



Admin control panel

Thank you

TransitBD — Group-5

Questions & Answers

github.com/Munem3/transit-ticketing-system